

AI-Powered Research & Report-Making

Achieve 10X Productivity in 1/10th of the Time with AI

BEFORE WE BEGIN...

1. If you have joined in from your phone, IT IS ABSOLUTELY **OKAY!**
2. As a part of today's session, **learn right now & practice later.** Recordings will be uploaded in the LMS after the session.
3. We will be focussing on delivering **maximum value per minute** in today's session. Please note all your questions and we will answer it in the **Q&A session** 🚀
4. Get ready to learn all about **Research and Report Making using AI**

The Problem with Traditional Research

Before AI, research and report writing were time-consuming and inefficient:

- ✗ **Endless Google Searches** – Manually reading multiple articles, papers, and reports.
- ✗ **Unstructured Information** – Copy-pasting disorganized notes into documents.
- ✗ **Time-Intensive Summarization** – Reading and extracting key insights manually.
- ✗ **Design & Formatting Hassles** – Struggling with PowerPoint & Excel visuals.
- 💡 AI changes the game by automating all of these processes, saving 10X the time!

Agenda for today's Session

- Introduction to AI-Powered Research
- AI Tools for Info Collection
- Optimizing Research Findings
- Designing Reports with AI
- Live Case Study
- Takeaways

Introduction to AI-Powered Research

AI-powered research accelerates data collection, summarization, and report creation. Tools like Perplexity, DeepSeek, and NotebookLM extract insights, summarize content, and cite sources instantly.

This automation makes research 10X faster, enhancing accuracy and efficiency, allowing professionals to focus on analysis, decision-making, and creating data-driven reports effortlessly.

AI Tools for Information Collection & Research

We will explore several AI tools by actually using them. Get ready for live demos – and I encourage you to follow along on your devices if you can!



12ft.io



Perplexity AI

The Answer Engine with Sources



Perplexity AI as an AI-driven answer engine or search assistant. Instead of just links, Perplexity gives you direct answers to your questions, with references. It's like Google + Wikipedia + ChatGPT combined.

Key Features:

- Web & Personal Data Search: "Merge information from both the web and your own files into one organized space"
- Cited Answers
- Multiple AI Models: You can select different AI brains (models) for answering. For instance, GPT-4 for creativity or DeepSeek R1 for deep reasoning
- The ability to switch models can yield better answers for different tasks.
- Collaborative Spaces

DeepSeek

Going Deeper in Search



DeepSeek (specifically the DeepSeek R1 model) is a state-of-the-art open-source language model known for advanced reasoning. It's making waves by tackling complex queries that other models struggle with.

Key Features:

- Exceptional at multi-step reasoning – great for tough analytical questions or when you need an explanation of how the answer was derived.
- Provides detailed reasoning traces (it will show its thought process), which is useful for transparency.
- Open-source origin means the community is testing and improving it rapidly.

Notebook LM

Your AI Research Notebook

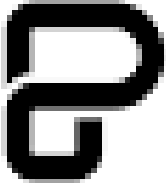


NotebookLM as Google's experimental AI research assistant (formerly known as Project Tailwind). It's like having an intelligent notebook that you can feed documents to and ask questions.

Key Features:

- Upload or import files (Docs, PDFs, Slides, URLs)
- Auto-Summary & Highlights
- This means it quickly distills long documents into the main ideas.
- Q&A with Sources: Ask it questions about your documents
- Great for extracting facts without losing where they came from.
- Idea Generation

Designing the report

 **PIKTOCHART**

upword*

 **Napkin**



Extracting and Verifying Information Quickly

- **Efficient Extraction:** Best practices for using AI to pull out exactly what you need from the sea of information:
 - **Use targeted questions:** e.g. “According to the 2022 report, what was the growth rate?” rather than “Tell me about the report.” Specificity helps AI pinpoint answers.
 - **Leverage summarization:** If you have a long article, ask the AI for a summary or even a specific detail (as some participants just did). You can also request “summarize this text and list any statistics with their sources” to extract data points.

Extracting and Verifying Information Quickly

- **Verification is Key:** AI is fast, but we must ensure accuracy:
- **Cross-Check Sources:** If an AI tool provides a source, click it. Does the source actually confirm the fact? Quick scanning of the original text can catch misinterpretations.
 - If no source is given, try a **second method:** ask another AI or do a traditional web search on the key fact. Consistency across answers increases confidence in the info.
 - Watch out for **AI-generated hallucinations** (fabricated info).

Smart Prompting Techniques for Precise Insights

Ask questions greatly influences the quality of AI's answers. A well-crafted prompt can save a lot of time by yielding the exact info or format you need on the first try.

Techniques & Examples:

- **Be Specific and Contextual:** Provide background or context in your question.
- **Ask for Output Format:** If you need a list, table, or summary, say so. "Give the answer in 3 bullet points" or "Provide a one-paragraph executive summary."
- **Use Follow-up Questions:** AI tools in chat format (ChatGPT, Bing Chat, etc.) remember context. You can drill down: "Now give me an example for point #2" or "Cite a source for that statistic." Iterative prompting refines the results without starting from scratch.
- **Incorporate Constraints:** For precision, you might add constraints like timeframe or source type. "Summarize recent (2021-2023) research on renewable energy from academic sources." This can focus the AI's attention.

Prompt Engineering Hacks

- Chain of thought prompting
- Multishot prompting
- Act as prompts, IQ prompts

Optimizing Research Findings with AI

Using AI to Summarize and Structure Information

Rapid Summaries: Emphasize that one of AI's superpowers is summarization. It can take a 20-page document or a long article and produce a concise summary in seconds.

- **Example:** “You could feed an AI a lengthy annual report, and ask for a 1-paragraph summary of the company’s financial performance — imagine getting the highlights without plowing through 50 pages!”
- Many AI tools (NotebookLM, ChatGPT, Claude, etc.) can summarize. ChatGPT can do it too if you paste text or give it a link (with browsing enabled or via plugins).

Optimizing Research Findings with AI

Outlining & Structuring: AI can help organize information logically:

- **Outline Generation:** Ask AI to generate an outline for your report. E.g. “Given these 5 bullet points (findings), draft a logical report outline with sections.” The AI might output a well-structured outline that you can directly use to write or present.
- **Categorizing Info:** If you have a list of mixed findings, you can prompt, “Group these points into themes.” AI might categorize insights into topics (for example, “Productivity Stats”, “Employee Sentiments”, “Challenges”).
- **Drafting Paragraphs:** You can even have AI expand an outline into prose. For instance, “Write a paragraph elaborating on point 2 of the outline.” This can kick-start your writing process.

Leveraging AI for Citations and Fact-Checking

Citations Made Easy: Perplexity and Bing Chat provide sources. Additionally, AI writing assistants can help format citations or suggest references when you draft text.

- For example, if you use a tool like SciSpace's AI assistant or Zotero's integration, they can recommend relevant papers and provide properly formatted citations (APA, MLA, etc.) for your report.
- Even ChatGPT, if instructed properly, can output a references section (though caution: it sometimes fabricates sources). One approach is using AI to find the info, then using reference management software to insert the actual citation.

Fact-Checking with AI

- **Ask the AI to show evidence:** e.g. “Can you provide a source for that data?” If the AI can search (like Bing), it might give a reference. If not, it’s a sign you should verify it yourself via a quick web search.
- **Cross-verify claims:** If a fact is important, run a separate query in a trusted source. For example, if the AI says “Project X cost \$5 million”, check Wikipedia or news articles independently to ensure that number is correct.
- **Use AI fact-checkers:** Some tools specialize in fact-checking (though nascent). They input a claim and output a true/false with evidence. While we might not demo one, know that the concept exists as AI tries to self-verify.

Fact checking hack

Reducing Manual Work

AI can handle the mundane but crucial part of reports – bibliography and reference lookup – much faster. Instead of manually collecting every URL or publication detail, you can have AI list out which sources you should cite for each claim.

- **Example:** After writing a draft, you could prompt, “Highlight any claims in my text that need citation.” The AI could underline statements that sound like facts and even suggest a source if it knows one.

Productivity Hacks for Organizing Research Data

Organize as You Go: Use AI not just for finding info, but for keeping it organized:

- **AI-Powered Notes:** Use tools like Notion with AI or Evernote with AI integrations to automatically categorize notes.
- **Semantic Search in Documents:** If you have a large collection of research docs, consider an AI tool that can search within all of them (Notebook LM's multi-document queries or even Microsoft Copilot in OneDrive).
- **Summarize Meetings/Interviews:** If your research involves meetings or interviews, AI transcription tools (like Otter.ai with AI summary) can transcribe and summarize key points.

Productivity Hacks for Organizing Research Data

Project Hub: Create an “AI research space” for each project:

- For example, in Perplexity Spaces, gather all relevant docs and Q&As in one place. Or in a shared Google Doc, continually ask ChatGPT (via copy-paste) to summarize new info. Having everything in one hub prevents siloed information.
- **Use cloud collaboration** – perhaps you and colleagues each use AI on different subtopics, then compile results in a shared folder. AI tools like Perplexity can act as that hub where everyone can see the collected intelligence.

Designing Reports with AI

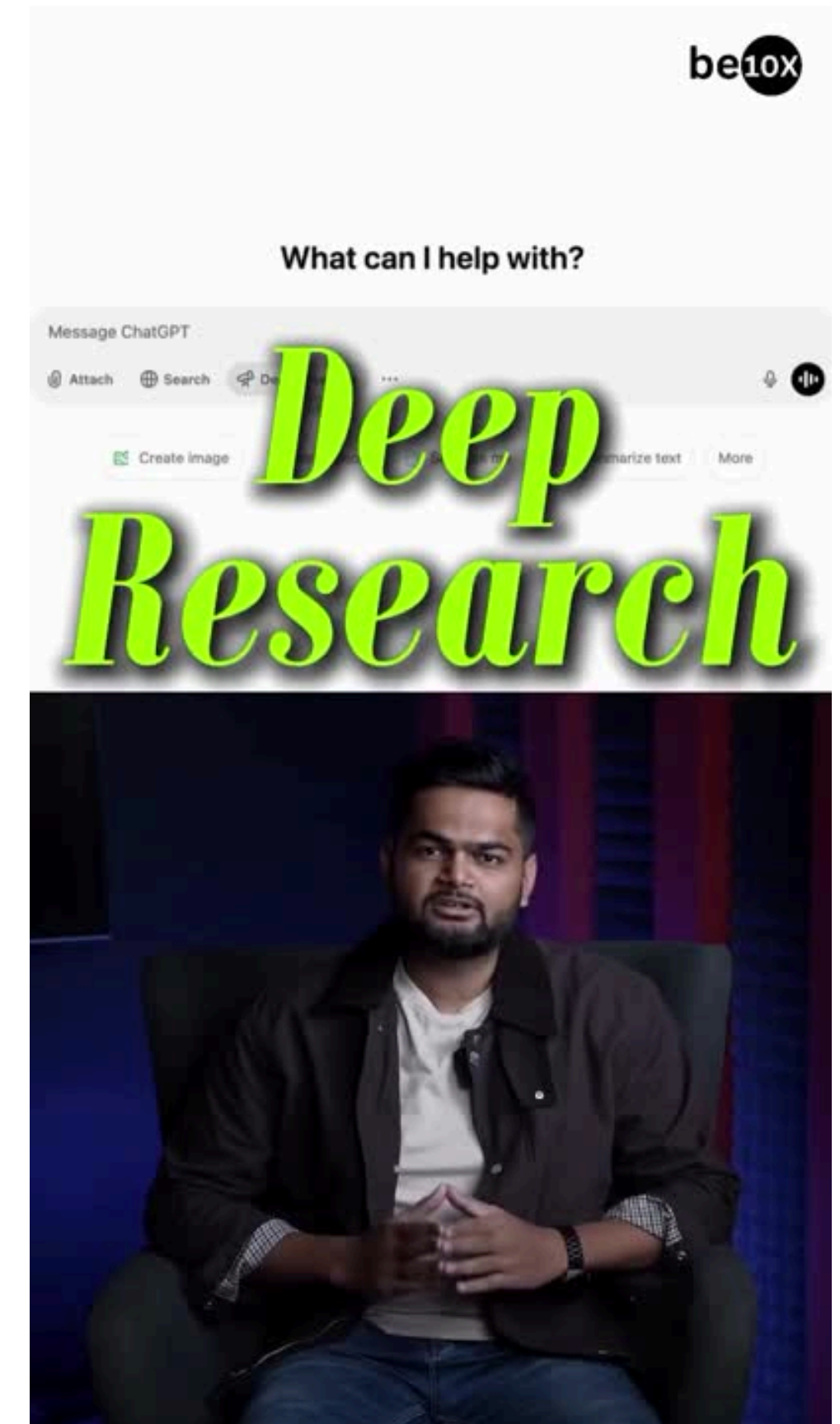
From Draft to Beautiful

AI-Powered Design Tools

Canva's Magic Design: Introduce Canva as a user-friendly design platform that now has powerful AI features.

- You describe what you need (e.g. “a 5-slide status report on Q4 sales, with a modern theme”), and Canva will produce a set of slide designs you can choose and edit.
- Canva can also suggest color schemes, images, and graphics based on your content.

OpenAI deep research



<https://www.instagram.com/be10x.in/>

Live Case Study: AI Workflow in Action

Bringing It All Together – From Blank Page to Report using AI

Research → Draft → Design

Step 1: Information Gathering Using AI

Objective:

- ✓ Use AI tools to gather credible and fast research.
- ✓ Extract key statistics, trends, and insights for reports.
- ✓ Identify credible sources and citations to maintain accuracy.

Key AI Tools for Research:

- 1 Perplexity AI** – Instant AI-powered search with sources.
- 2 DeepSeek AI** – Deeper, more analytical research findings.
- 3 NotebookLM** – AI that digests PDFs and documents for insights.
- 4 Bing AI / Google AI** – Real-time web search with citations.

Question → AI Research → Extracted Insights → Ready for Report

Step 2: Organizing AI Findings into a Coherent Structure

Objective:

- ✅ Convert AI-researched data into clear summaries & categories.
- ✅ Create an outline for a structured report.
- ✅ Identify trends, patterns, and key insights from research.

How AI Helps:

- 💡 **NotebookLM & ChatGPT** – Summarize long documents into key takeaways.
- 💡 **AI Outlining Tools** – Auto-generate structured report frameworks.
- 💡 **Perplexity AI** – Refines raw research into concise sections.

Let's Create a Report Using AI!

Instructions:

- 1 Step 1:** Use Perplexity AI to find 2 key insights on remote work.
- 2 Step 2:** Use ChatGPT/NotebookLM to summarize and structure findings.
- 3 Step 3:** Use Canva AI to create a simple 3-slide report.

Expected Outcome:

- ✓ AI-generated research-backed report in 15 minutes!
- ✓ Practical experience with AI-powered research & design.
- ✓ Hands-on application of all concepts learned so far.

AI Productivity Hacks – What to Remember!

Top 5 AI Hacks:

- ✓ **Always Ask AI First** – Saves hours on research.
- ✓ **Be Specific with Prompts** – Better prompts = better outputs.
- ✓ **Use AI for Outlines** – AI drafts help structure content fast.
- ✓ **Verify Information** – Always check AI-generated citations.
- ✓ **Leverage AI for Design** – Reports look professional in minutes!

Your AI Toolkit for Research & Reports

Best AI Tools Recap:

- ✓ **Research:** Perplexity AI, DeepSeek, NotebookLM.
- ✓ **Summarization & Writing:** ChatGPT, QuillBot, PaperPal.
- ✓ **Report Design:** Canva AI, Beautiful.AI, Piktochart.
- ✓ **Organization & Productivity:** Notion AI, Otter.ai.

Implementing AI at Work

- ✓ **Start small** – Try AI for one research task this week.
- ✓ Practice summarizing and outlining with **AI**.
- ✓ **Use AI tools** for real projects & share success stories!

**Create a detailed report
on impact of GenAI on the
industry that you are
currently working in**

Rate this session

Thank You

Thank you for joining! Stay connected & explore AI.